

1 Connection of HYDRA Time Management to SAP R/3 HR

1.1 Summary

This document describes the data exchange on the logical level between HYDRA Personnel Time Management and SAP-HR.

The connection of the HYDRA Personnel Time Management module (PZW) to SAP-HR is considered an “intelligent subsystem”. In this context, evaluated results are forwarded to SAP and integrated in SAP payroll accounting.

Communication is performed using SAP-I-Docs. Consequently, “ALE customizing” is required for the implementation of interfaces to establish the data connection between SAP and HYDRA.

1.2 Download of the HR master

HR master data can either be transferred using the SAP standard interface as a part of HR-PDC or by the HYDRA standard interface.

The HR-PDC interface provides the advantage that it requires only little customizing and, as a result, is immediately available. However, it does not include all HR master data that are required for a reasonable operation of HYDRA. Consequently, manual rework is required in most cases.

The HYDRA interface provides access to all fields of the HR master. However, a customer-specific function module is required in SAP, which is able to deal with the interface format that is specific to HYDRA.

The HR master interface by HR-PDC is described in the document entitled “HYDRA Interfacing module to SAP HR using HR-PDC” (SAP-HRPDC_30.pdf).

The standard HR master interface of HYDRA is described in the document entitled “Interface Wage and Salary Programs” (EIS-LUG_81.pdf).

1.3 Download of account balances

As time management is performed in HYDRA if SAP-HRZW is in use, HYDRA is also the guiding system with respect to time and leave accounts. An interface is not planned for this reason.

1.4 Upload of results

1.4.1 Determination and customizing of the source system

SOURCE_SYS

The target SAP system is included as SOURCE_SYS in the transferred data of many upload interfaces.

The SAP source system of the person is transferred and stored in the 16th individually configurable information field of the HYDRA HR master, once HR masters have been loaded in the SAP format using HR-PDC in HYDRA. No further configuration will be necessary.

If the HR master is maintained in another manner the entry will not be available and the source system for the upload can be determined using the HYDRA ALE configuration. For this purpose, the source system of an active logical SAP system is read, which can be set by an INI configuration in HYDRA.

The source system is determined based on the following rule and priority. The other rules will not apply, provided that a source system could be determined using the listed rules in the below-mentioned order.

1. Entry in the information field 16 of the HR master

If an entry is included in this field it will be interpreted as source system for the upload.

2. Using the logical system from the INI configuration for the personnel number

A logical SAP system is specified for the personnel number using the INI configuration:

Name of the INI file	"HR-LOGSYS"
Section	required logical system
Key	"PNR"
Value	Personnel number of the requested person

The active source system of the logical system is determined.

3. Using the logical system from the INI configuration for the company

A logical SAP system is indicated for the company from the HR master using the INI configuration:

Name of the INI file	"HR-LOGSYS"
Section	required logical system
Key	"FIR"
Value	Company

The active source system of the logical system is determined.

4. Using the logical system from the INI configuration, default entry

An entry to generally specify a logical SAP system can be made using the INI configuration:

Name of the INI file	"HR-LOGSYS"
Section	required logical system
Key	"ALL"
Value	"Y"

The active source system of the logical system is determined.

5. Default determination

The active source system of the logical system "SAP" is determined.

The field remains empty if no source system could be determined by means of the listed rules.

1.4.2 Data upload by ALE (customizing)

SAP does not support upload requests for the message types of the interfacing module SAP-HRZW.

To transfer the data provided in HYDRA outbound transaction to an SAP system, it is the easiest way to configure the upload in HYDRA and to run it cyclically.

The cyclic upload is configured in the HYDRA MLE distribution model and the [HYDRA-Scheduler](#) and is normally carried out by MPDV staff:

A new entry is to be created in the Scheduler indicating the following details:

Type:	I - Interval
Type:	C – customer entry
Product key:	SAP-HRZW
Active	☒
License key:	
HYDRA user:	
Command (Windows) :	sh.exe ./hysapupl.scr /UPLSEGNAM=<Segment name>
(Unix) :	sh ./hysapupl.scr /UPLSEGNAM=<Segment name>
e.g. :	sh.exe ./hysapupl.scr /UPLSEGNAM=E2BP7012_2001
Interval:	00:30 Hours:Minutes
From:	00:00 a.m./p.m.
Until:	00:00 a.m./p.m.

This exemplary entry makes sure that the segments of the specified type provided in open outbound transactions are physically transferred to the SAP system specified in the MLE distribution model every 30 minutes.

1.4.3 Upload of wage types

Wage types are transferred to SAP-HR using the info type 2010.

1.4.3.1 Trigger the upload

The HYDRA function [interface to payroll accounting](#) triggers the upload of the SAP-I-Doc. The wage types accumulated for the settlement period are transferred to SAP-HR, once data have been requested by clicking the button with the green arrow. The data are also displayed on the screen.



It is not planned to cancel data that have already been transferred beforehand by this interface. If the upload is triggered once more data will again be transferred to SAP. The concerned data have to be cancelled manually in SAP if data are required to be transferred once more.

1.4.3.2 Data record structured of the wage type interface

The SAP Bapi "external wage types in info type 2010" is used.

Message type: REM_SPEC_WITH_COST
IDoc type: REM_SPEC_WITH_COST01
Segments: E2BP7012_2001

Field name	Type	L	D	Pos	Description	Used in HYDRA
EXTSYSTEM	CHAR	10		1	Logical system of the original document	Yes
EXTAPPLICATION	CHAR	5		11	External application	"ZHYD" please see the below notes
EXTDOCUMENTNO	CHAR	20		16	Document number	Unique ID of the posting, time stamp + data record number
REVERSED	CHAR	1		36	ID: Data record has been cancelled	Empty
CUSTOMER_FIELD	CHAR	40		37	Individual customer field	Empty
EMPLOYEEENUMBER	NUMC	8		77	Personnel number	Personnel number
VALIDITYDATE	DATS			85	Validity date	Posting date (last day of the month; restricted to the date of leaving the company)
LOCKINDICATOR	CHAR	1		93	Blocking flag for HR master record	Empty
WAGETYPE	CHAR	4		94	Wage type	Wage type according to HYDRA
NO_OF_HOURS	DEC	7	2	98	Number of hours for payment documents	Empty
NUMBER	DEC	7	2	107	Number for each time unit for payment documents	Duration (monthly total)
TIME_UNIT	CHAR	3		116	Unit of time/measure	001 (hours) 010 (days) are transferred for wage types with average type "T".
TIME_UNIT_ISO	CHAR	3		119	ISO code unit of measure	Empty

Field name	Type	L	D	Pos	Description	Used in HYDRA
AMOUNT	DEC	23	4	122	Amount	Empty
OT_COMP_TYPE	CHAR	1		147	Compensation type of overtime	Empty
WORKTAXAREA	CHAR	4		148	Tax area: workplace	Empty
PAYSCALEGROUP	CHAR	8		152	Wage group	Empty
PAYSCALELEVEL	CHAR	2		160	Pay level	Empty
BONUSTYPE	CHAR	2		162	Bonus number	Empty
BONUSVALUE	NUMC	4		164	Bonus ID	Empty
POSITION	NUMC	8		168	Planned position	Empty
VALUATION_BASIS	DEC	23	4	176	Valuation basis	Empty
CURRENCY	CUKY			201	Currency key	Empty
CURRENCY_ISO	CHAR	3		206	ISO code currency	Empty
EXTRA_PAY_INDIC	CHAR	1		209	Extra pay indicator	Empty
COMP_CODE	CHAR	4		210	Company code	Empty
BUS_AREA	CHAR	4		214	Business area	Empty
COSTCENTER	CHAR	10		218	Cost center	Cost center from wage type posting
ACTTYPE	CHAR	6		228	Activity type	Empty
ORDERID	CHAR	12		234	Order number	Empty
COST_OBJ	CHAR	12		246	Cost carrier	Empty
WBS_ELEMENT	CHAR	24		258	Planned element of project structure (PSP element)	Empty
NETWORK	CHAR	12		282	Network plan number for accounting	Empty
ACTIVITY	CHAR	4		294	Operation number	Empty
SALES_ORD	CHAR	10		298	Sales order number	Empty
S_ORD_ITEM	NUMC	6		308	Item number in sales order	Empty
CO_BUSPROC	CHAR	12		314	Business process	Empty
SERVICE_TYPE	CHAR	2		326	Service type (PSG)	Empty
SERVICE_CATEGORY	CHAR	2		328	Sub-service type (PSG)	Empty

Please note for the field **EXTAPPLICATION**:

The external application "ZHYD" has to be created in the SAP system:



Transaction SPRO → Personnel Time Management → Integrate Time Management with other applications → Define external application for the integration with Personnel Time Management. It is a transportable customizing that is created in the development system and has to be transported.

The end of the month is transferred as the validity date (VALIDITYDATE) for the upload. The date of leaving is used for employees who left the company during the month.

Data are compressed by month, wage type and cost center for each person. The cost center is transferred with every data record, even if does not differ from the master cost center.

1.4.4 Upload of absences

The transfer of absences to SAP-HR is not enabled automatically, but has to be set while customizing the interface.

Which absences are to be uploaded can be configured in Master data → Labor time → Control of absences (transaction code ABSE) by the button "Upload to payroll accounting". The absence reason can be defined for SAP in the below field "absence reason", provided that this reason deviates from the number of the corresponding absence payment.

1.4.4.1 Trigger the upload

The HYDRA function [interface to payroll accounting](#) triggers the upload of the SAP-I-Doc along with the upload of wage types. Absence times are written into the HYDRA outbound transactions, once data have been requested using the button with the green arrow.



It is not planned to cancel data that have already been transferred beforehand by this interface. If the upload is triggered once more data will again be transferred to SAP. The concerned data have to be cancelled manually beforehand in SAP if this repeated transfer of data is required.

Data can be displayed within the open data segments of [MLE outbound transactions](#) as long as they have not yet been transferred to SAP. Then they can directly be displayed within the outbound transactions.

1.4.4.2 Data record structure of absence interface

Absences are transferred to SAP-HR using the info type 2001.

Message type: ATT_ABS
IDoc type: ATT_ABS01
Segments: E2BP7011_1000

Field name	Type	L	D	Pos	Description	Used in HYDRA
EXTSYSTEM	CHAR	10		1	Logical system of the original document	Yes
EXTAPPLICATION	CHAR	5		11	External application	Fixed "ZHYD" Please see the below note
EXTDOCUMENTNO	CHAR	20		16	Document number	Internal key
REVERSED	CHAR	1		36	Flag: data record has been cancelled	This interface does not provide for cancellation records
CUSTOMER_FIELD	CHAR	40		37	Individual customer field	Empty
EMPLOYEENUMBER	NUMC	8		77	Personnel number	Personnel number
FROM_DATE	DATS	8		85	Beginning of validity	Start date of the absence, restricted to the beginning of the month
TO_DATE	DATS	8		93	End of validity	End date of the absence, restricted to the date of leaving the company or the end of the month
LOCKINDICATOR	CHAR	1		1	Blocking flag	Remains empty

Field name	Type	L	D	Pos	Description	Used in HYDRA
ABS_ATT_TYPE	CHAR	4		102	Start of presence or absence	Number of the absence payment in HYDRA or the absence reason configured in "absence processing"
START_TIME	CHAR	6		106	Start time	Empty
END_TIME	CHAR	6		112	End time	Empty
ABS_ATT_HOURS	DEC	7	2	118	Hours of attendance/absences	For half days of absence: net hours according to absence clocking (breaks are deducted) otherwise: 0,0
ALL_DAY_FLAG	CHAR	1		127	Data record applies for the whole day	Remains empty
PREVIOUS_DAY	CHAR	1		128	Previous day flag	Remains empty
OT_COMP_TYPE	CHAR	1		129	Compensation type of overtime	Remains empty
WORKTAXAREA	CHAR	4		130	Tax area: workplace	Remains empty
PAYSCALEGROUP	CHAR	8		134	Wage group	Remains empty
PAYSCALELEVEL	CHAR	2		142	Pay level	Remains empty
BONUSTYPE	CHAR	2		144	Bonus number	Remains empty
BONUSVALUE	NUMC	4		146	Bonus ID	0
POSITION	NUMC	8		150	Planned position	0
VALUATION_BASIS	DEC	23	4	158	Valuation basis	0,0 (without algebraic sign)
CURRENCY	CUKY	5		183	Currency key	Remains empty
CURRENCY_ISO	CHAR	3		188	ISO code currency	Remains empty
EXTRA_PAY_INDIC	CHAR	1		191	Extra pay indicator	Remains empty

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The interface includes the compensated (past) and not the planned (future) absences. When it comes to full-time absences, the periods are transferred coherently as one data record including weekends and other days off and not as individual, subsequent absences. Absences are uploaded for each [settlement period](#). Two separated absences are transferred if an absence period extends over two months (or several absences if several months are affected).

The dates for the beginning and end of the validity are identical for **part-time absences**. The actual absence determined in HYDRA is entered in the field ABS_ATT_HOURS (hours of presence/absence). The fields START_TIME (start time) und END_TIME (end time) remain empty. Up to two part-time absences are processed by the interface per day.